

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To solve, first calculate the normal download time for the whole file, then the time for the 40% that needs to be red one, and finally add the 20 minutes for the Windows update restart. $200 \text{ GB} / 2 \text{ GB per minute} = 100 \text{ minutes}$ for the whole file at normal speed. $40\% \text{ of } 200 \text{ GB} = 80 \text{ GB}$. $80 \text{ GB} / 2 \text{ GB per minute} = 40 \text{ minutes}$ to download 80 GB. Then 40 minutes to red ownload, and 20 minutes for Windows to update, making the total $100 + 40 + 20 = 160 \text{ minutes}$. </think><answer>160</answer>

